

Real-time Rock Paper Scissors Game Using OpenCV and TensorFlow

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Introduction

The classic Rock Paper Scissors game is given a twist with this project, which uses real-time hand gesture recognition. This system employs computer vision techniques, using Python, OpenCV, and TensorFlow, to classify hand gestures captured by a webcam, allowing a user to play the game against the computer.

Motivation and Existing Challenges

Hand gesture recognition is a valuable area of research with applications in human-computer interaction, virtual reality, and sign language interpretation.

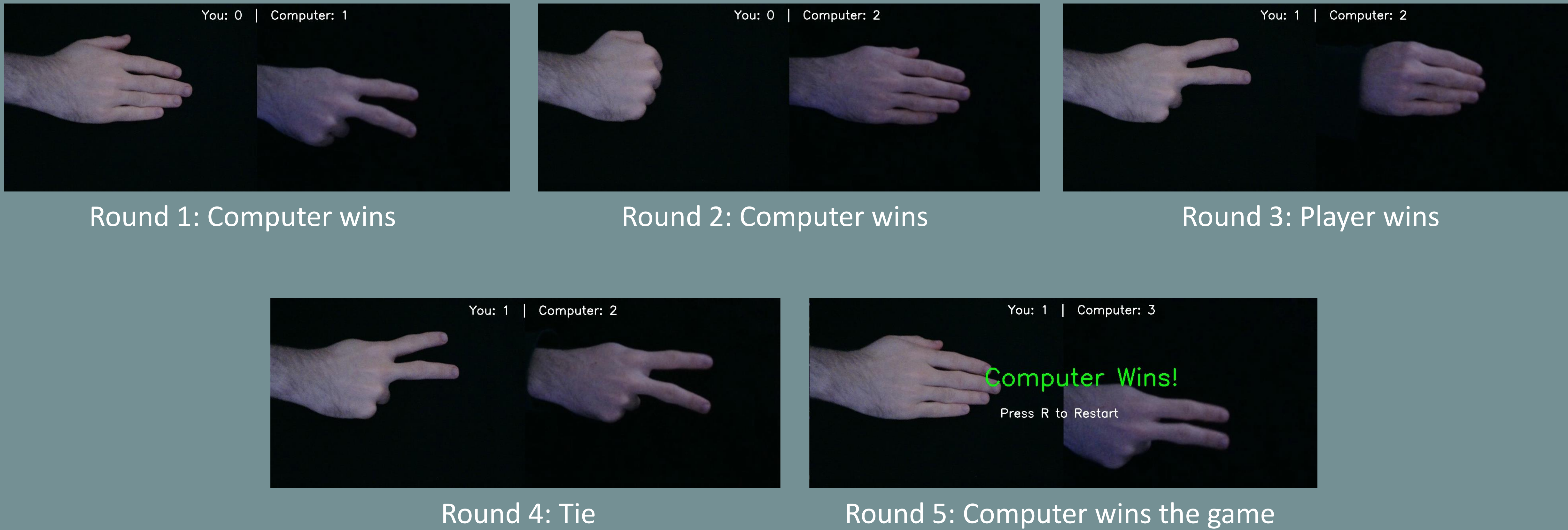
- However, real-time hand gesture recognition presents several challenges:
- Variation in hand shapes and sizes
 - Lighting conditions
 - Background clutter
 - Speed and accuracy trade-offs

My Contributions

- This project contributes to the field by:
- Developing a real-time Rock Paper Scissors game using hand gesture recognition.
 - Implementing a CNN model for accurate hand gesture classification.
 - Creating a functional and engaging game experience that demonstrates the potential of real-time hand gesture recognition.

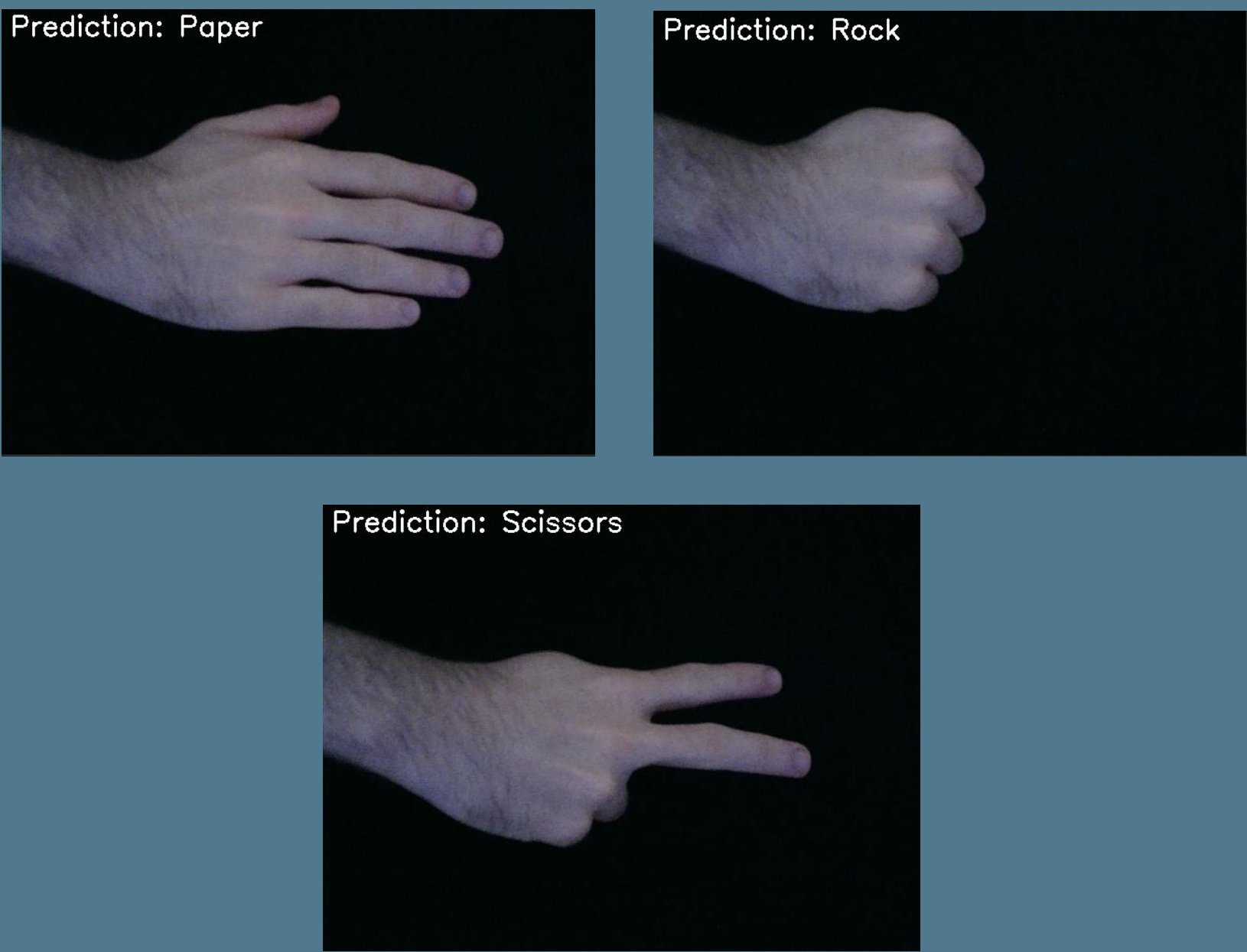
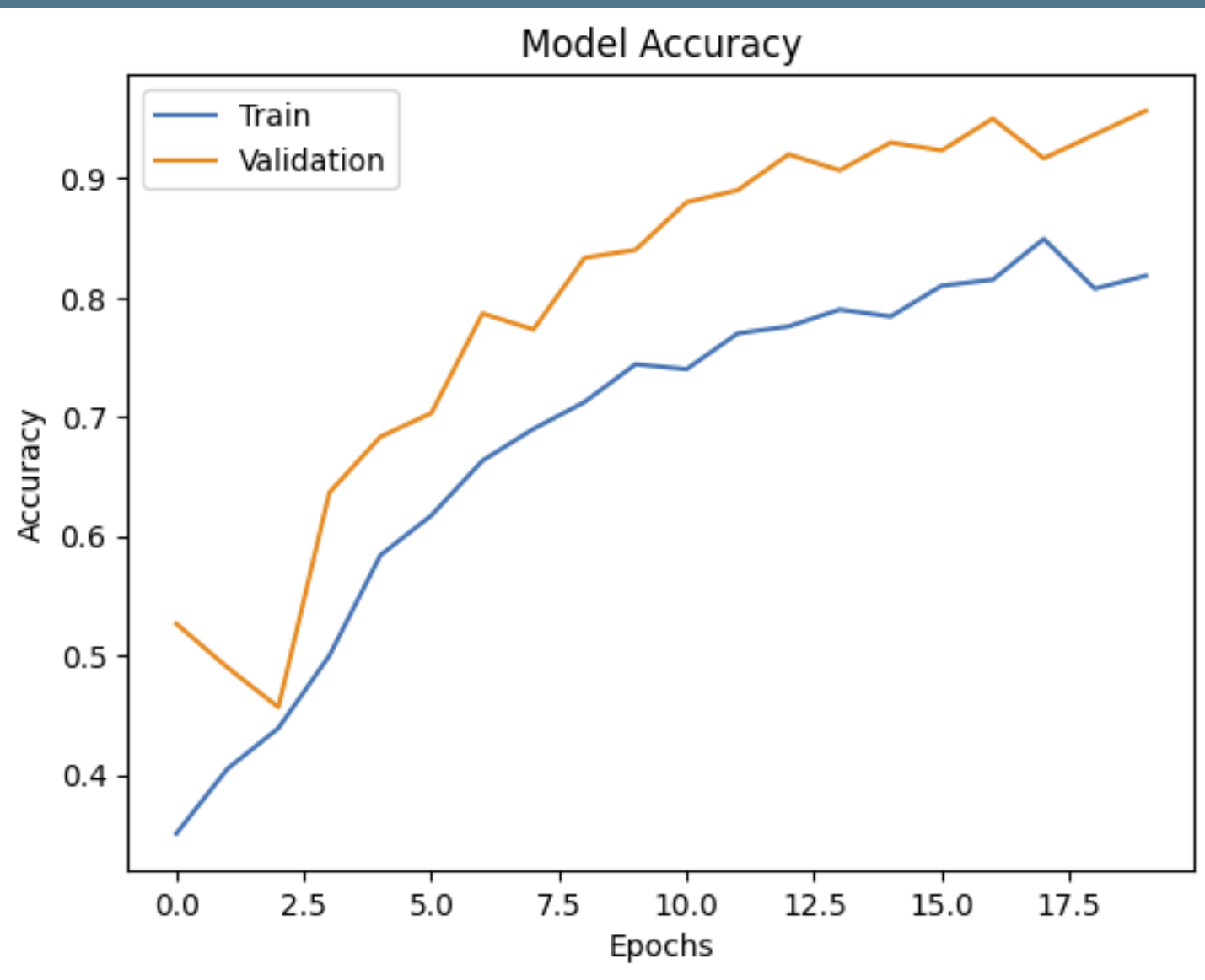
Methodology

- CNN Model:**
 - A Convolutional Neural Network (CNN) was designed to classify hand gestures into three categories: Rock, Paper, and Scissors. The CNN architecture comprises three convolutional layers (32, 64, 128 filters, 3x3 kernel, ReLU activation), max pooling (2x2), two fully connected layers (128 and 3 units, ReLU activation on the first), and a softmax output layer.
 - The model was trained on a dataset of hand gesture images containing 500 images from each class. The dataset was split into training and validation sets. The model was trained with a learning rate of 0.001 and the cross-entropy loss function. The model was trained for 20 epochs with a batch size of 32.
- Image Processing:**
 - The webcam feed is captured using OpenCV.
 - Each frame is processed to extract the hand region and prepare it for the CNN. Preprocessing steps include resizing the frame to 320x240, converting the image to RGB color space and normalizing the pixel values in range of 0 to 1.
- Game Logic:**
 - The webcam captures the player's hand gesture.
 - The CNN model predicts the player's gesture.
 - The computer randomly selects its move.
 - The winner is determined based on the rules of Rock Paper Scissors.
 - The game state (scores, results) is displayed in real-time.



Results

The game demonstrates robust real-time hand gesture recognition. The CNN model achieves an accuracy of 0.95 in classifying hand gestures.



Conclusion

This project demonstrates the feasibility of using computer vision and deep learning for real-time hand gesture recognition in an interactive application. The Rock Paper Scissors game provides an engaging example of this technology, which has the potential to be extended to a wide range of other applications.